

Johann Sebastian Quenta Raygada

- Ph.D. Student in Theoretical Particle Physics

Education

Scuola Internazionale Superiore di Studi Avanzati (SISSA) <i>Ph.D. in Theoretical Particle Physics</i> <i>Trieste, Italy</i>	2023 - 2027
Université de Bourgogne (uB) <i>M2 Math4Phys Master's Program in Mathematical Physics</i> <i>Dijon, France</i> • Thesis: <i>Conformal Field Theories of Generalized Parafermions</i> ↗	2022 - 2023
The Abdus Salam International Centre for Theoretical Physics (ICTP) <i>Diploma Programme in High Energy, Cosmology and Astroparticle Physics</i> <i>Trieste, Italy</i> • Dissertation: <i>Generalized Global Symmetries</i> ↗	2021 - 2022
Pontificia Universidad Católica del Perú (PUCP) <i>B.Sc. in Physics</i> <i>Lima, Perú</i> • Dissertation: <i>Stability analysis of chemical fronts in exothermic reactions</i> ↗	2016 - 2020

Research Experience

Chern-Simons theory for non-semisimple gauge groups <i>With Pavel Putrov (ICTP), Thomas Nicosanti (SISSA)</i>	In progress
Minimal modular extensions of symmetry categories <i>With Rajath Radhakrishnan (Oxford)</i>	In progress
SU(2) Chern-Simons invariants from discrete gauge theories <i>With Pavel Putrov (ICTP), Thomas Nicosanti (SISSA)</i>	2024 - 2025
Conformal field theories of generalized parafermions <i>With Taro Kimura (uB)</i>	2022 - 2023
Analysis of chemical front stability in exothermic reactions <i>With Desiderio Vasquez (PUCP)</i>	2019 - 2021

Publications

<i>Approximating SU(2) Chern-Simons theory by finite group gauge theories</i> <i>Thomas Nicosanti, Pavel Putrov, Johann Quenta-Raygada</i> <i>arXiv preprint: 2512.11704</i> ↗	Dec 12, 2025
<i>Thermal and compositional driven convection in thin reaction fronts</i> <i>Johann Quenta, Desiderio Vasquez</i> <i>In Physical Review E</i> ↗	Mar 28, 2024

Conferences and Schools

<i>Physics and Mathematics of Generalized Symmetry in QFT</i> <i>Kavli Institute for the Physics and Mathematics of the Universe</i>	Aug 2026
<i>Global Categorical Symmetries 2026 (School and Conference)</i> <i>Institut Henri Poincaré</i>	Jun 2026
<i>2025 SCGCS School - Topological Symmetries and Defects in Quantum Field Theory</i> <i>SwissMAP Research Station</i>	Jun 2025
<i>Hot Topics Conference on Quantum Topology</i> <i>Max Planck Institute for Mathematics</i>	May 2025
<i>Symmetries and Anomalies: A Modern Take</i> <i>Institut des Hautes Études Scientifiques</i>	Jun - Jul 2024
<i>Spring School on Superstring Theory and Related Topics</i> <i>The Abdus Salam International Centre for Theoretical Physics</i>	2022, 2023, 2025 & 2026
<i>Generalized Cohomology and Physics</i> <i>The Abdus Salam International Centre for Theoretical Physics</i>	Nov 2021

Awards and Distinctions

Joint ICTP/SISSA Ph.D. Fellowship <i>SISSA and The Abdus Salam International Centre for Theoretical Physics</i>	2023 - 2027
Math4Phys M2 Scholarship <i>Institut de Mathématiques de Bourgogne</i>	2022 - 2023
ICTP Postgraduate Diploma Programme Fellowship <i>The Abdus Salam International Centre for Theoretical Physics</i>	2021 - 2022

Teaching Experience

Teaching Assistant for <i>General Relativity I</i> <i>ICTP Postgraduate Diploma Programme in High Energy Physics</i>	2025
Teaching Assistant for <i>Quantum Field Theory I</i> <i>Physics Latam & ICTP Physics Without Frontiers</i>	2023
Teaching Assistant for <i>Electromagnetic Field Theory</i> <i>PUCP Science and Engineering Faculty</i>	2021
Teaching Assistant for <i>Fundamentals of Physics, Physics 1, Calculus 1</i> <i>PUCP General Science Studies Department</i>	2019 - 2021